

Offline Technical Task — Backend Lead

Deadline: September 20

Submit to: aviranmz@gmail.com

Email subject: Backend Lead - Offline Task - Oleksandr Shapovalov

Time budget

90 minutes. That is the expected time for a complete v1 of this brief.

If you are not finished at 90 minutes, **stop**, write what you would do next, and submit. **Going past 2 hours is a negative signal** for this role. A small working slice beats an unfinished platform.

Suggested split: 10 min read + assumptions, 50 min build + tests, 15 min README, 15 min deploy.

Comment Moderation API

A community product receives short user comments. They need a backend that:

1. Accepts a comment
2. Moderates it with a **replaceable AI provider** (decision, reason, one-line suggested reply to the user)
3. Stores the result
4. Lets them fetch the moderation by id

You are the **Backend Lead** delivering **v1** from zero. There is no existing codebase.

You may use **any language, framework, and database**. Node.js / TypeScript is a natural fit for this role but is not required.

AI coding tools are allowed (Cursor, Claude, Copilot, ChatGPT, etc.). We evaluate **your** judgment and ownership — not how much code a model can emit.

Default AI implementation is a mock. A real LLM is optional and must use the **same interface**. The reviewer must be able to run the service **without an API key**.

Read this brief yourself before you start.

Build (this is the whole task)

1. POST a comment with idempotencyKey, text, and optional userId
2. Call an AiModerationProvider (name yours if you prefer) that returns { decision, reason, suggestedReply }
 - decision is one of allow | flag | block
3. Persist the comment and moderation. Store which provider/model produced the result
4. GET by id
5. Health check

6. The same `idempotencyKey` must **not** call the provider again
7. Tests: happy path, duplicate key, invalid input

Keep the implementation **small**.

Out of scope: frontend, mobile apps, queues, Kafka, Saga, Kubernetes, authentication product, requiring a real OpenAI/Anthropic key, webhooks, blockchain, extra microservices.

Synthetic text only. No real user data.

Deliverables

Submit **one email** with:

1. **Git repository** — meaningful commit history (not a single “init” commit), `.env.example`, tests in repo
2. **README** — assumptions, architecture, API, data model, how to run tests, idempotency, trade-offs, **minutes you actually spent**, and an AI usage log if you used AI
3. **Runnable service** — public HTTPS URL with a health check **preferred**. If deploy does not fit the 90-minute budget, submit Docker Compose that starts the API (and database if you use one) and say so in the README.

Include your name and honest minutes spent in the email.

AI usage log

If you used AI, add this section to your README:

`## AI usage log`

Step	Tool	What I asked AI to do	What I kept	What I changed/rejected	Why
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`**Parts written without AI:**`

`**How I verified AI output:**`

Be honest. We will ask about it in the interview.

Technical interview

We will have read your repository. The session covers the provider interface, idempotency, what you cut to stay inside 90 minutes, tests, and how you would add a real LLM in v2 without rewriting the service.

Questions

Email aviranmz@gmail.com if blocked on an ambiguity. Reasonable documented assumptions are preferred over waiting.

iForAI — Backend Lead hiring — Oleksandr Shapovalov